# Area 6: Supportive Knowledge

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

## Contents

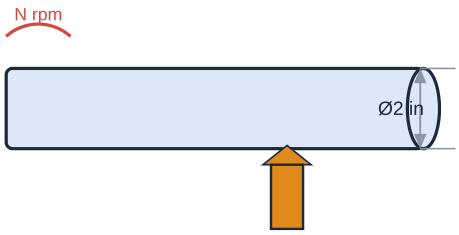
6.A Manufacturing Methods
6.B Fits and Tolerances
6.C Codes and Standards
6.D Computational Methods
6.E Instrumentation, Testing, Inspection, and Quality
6.F Chemical Processes (Corrosion, Coatings, Treatments)

# 6.A Manufacturing Methods

## 1. Key concepts

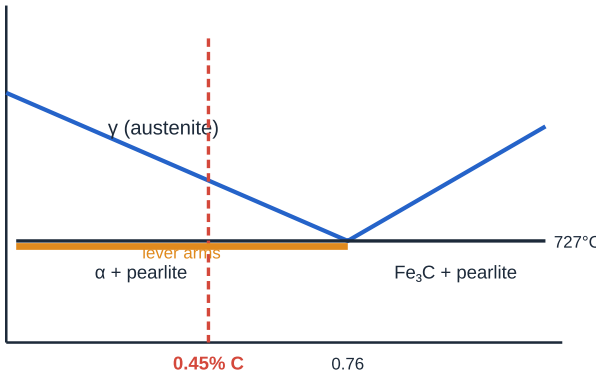
- Machining: spindle speed N = 12V/(πD) (V in sfm, D in in). Feed rate = f·N. Time = L/(fN). Taylor tool life $VT^n = C$.
- Process choice: casting (complex shapes, lower strength), forging (good grain flow, fatigue strength), machining (precision), additive (complex shapes, low volume), thread rolling (cold work, better fatigue).
- Heat treatment: annealing (soft), normalizing, quench and temper, case hardening (carburizing or nitriding).

## Diagrams




*V = 500 sfm; N = 12V/(πD)*

Turning: cutting speed V, spindle speed N, feed f.




Heat treatment depends on carbon content and temperature.

## 2. Essential formulas

| Name | Formula | Variables / units |
|---|---|---|
| Spindle speed | $N = 12V/(\pi D)$ | |
| Cutting time | $t = L/(fN)$ | |
| Taylor | $VT^n = C$ | |
| MRR (turning) | $MRR = 12V \cdot f \cdot d$ | $in^3/min$ |

## 3. Handbook search keywords

cutting speed · Taylor tool life · machining time · manufacturing processes

## 4. Worked examples

**Example 1: Taylor tool life**

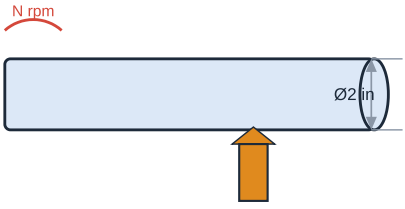
n = 0.25, C = 400. Find T at V = 300 ft/min.

Handbook search: **Taylor tool life**

1. T = (400/300)^(1/0.25) = 3.16 min.

**Answer:** 3.16 min

Check your answer: A faster cutting speed must give a shorter tool life.

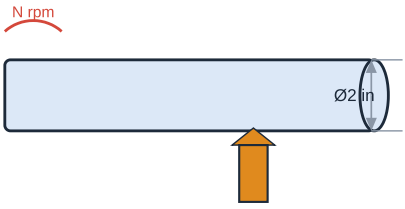


**Try it yourself: Spindle speed**
Turn a Ø2.0 in bar at 500 sfm. Find N.
Handbook search: **cutting speed**

1. N = 12V/(πD) = 12 × 500/(π × 2) = 955 rpm.

**Answer:** 955 rpm

Check your answer: The factor 12 converts feet to inches. Forgetting it gives an answer 12× too small.



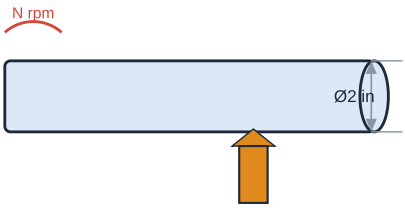
*V = 500 sfm; N = 12V/(πD)*

**Example 3: Cutting time**
A 10 in pass at f = 0.012 in/rev and 400 rpm. Find the time.
Handbook search: **machining time**

1. Feed rate = fN = 4.8 in/min.
2. t = L/(fN) = 2.08 min.

**Answer:** 2.08 min

Check your answer: Add approach and overtravel in real jobs.




## 5. Common traps

- Leaving out the 12 in N = 12V/πD.
- Inverting the exponent in Taylor (use 1/n).

# 6.A Manufacturing Methods: one-page cheat summary

► $N = 12V/\pi D$

► $t = L/fN$

► $T = (C/V)^{(1/n)}$

## Formulas

| | |
|---|---|
| Spindle speed | $N = 12V/(\pi D)$ |
| Cutting time | $t = L/(fN)$ |
| Taylor | $VT^n = C$ |
| MRR (turning) | $MRR = 12V \cdot f \cdot d$ |

## Search keywords

cutting speed, Taylor tool life, machining time, manufacturing processes

## Top traps

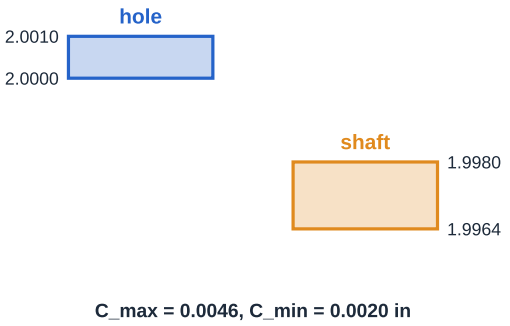
- Leaving out the 12 in N = 12V/πD.
- Inverting the exponent in Taylor (use 1/n).

# 6.B Fits and Tolerances

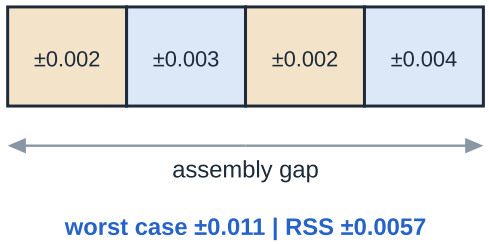
## 1. Key concepts

- Hole-basis system: the hole lower limit equals basic size (H). Max clearance = largest hole − smallest shaft. Min clearance (or max interference) = smallest hole − largest shaft.
- ANSI B4.1 classes: RC running, LC locational clearance, LT transition, LN locational interference, FN force/shrink. ISO: H7/g6 sliding, H7/h6 locational, H7/p6 interference, H7/s6 drive.
- Tolerance stack: worst case $T = \Sigma T_i$; statistical (RSS) $T = \sqrt{\Sigma T_i^2}$.

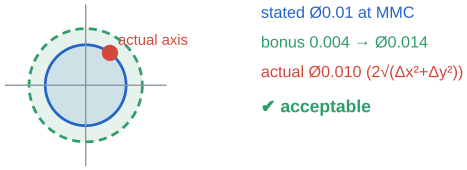
## Diagrams




Hole and shaft tolerance zones: max and min clearance.




Worst-case vs RSS stack-up.




Position tolerance zone.

## 2. Essential formulas

| Name | Formula | Variables / units |
|---|---|---|
| Max clearance | $C_{max} = hole_{max} - shaft_{min}$ | |
| Min clearance | $C_{min} = hole_{min} - shaft_{max}$ (negative = interference) | |
| Worst case | $T = \Sigma t_i$ | |
| RSS | $T = \sqrt{\Sigma t_i^2}$ | |

## 3. Handbook search keywords

limits and fits · ANSI B4.1 · ISO 286 · tolerance stack-up RSS

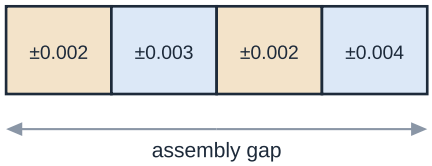
## 4. Worked examples

**Example 1: Stack-up**
Four parts at ±0.002, ±0.003, ±0.002, ±0.004 in. Find the worst-case and RSS tolerances.
Handbook search: **tolerance stack up**

1. Worst case = ±0.011 in.
2. RSS = $\sqrt{(4 + 9 + 4 + 16)} \times 10^{-3}$ = ±0.0057 in.

**Answer:** ±0.011 (WC), ±0.0057 (RSS)

Check your answer: RSS is always less than or equal to worst case.



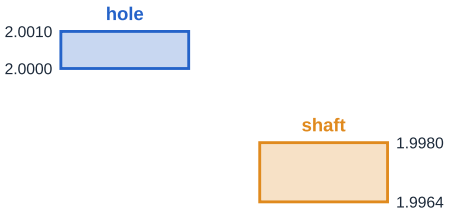

**worst case ±0.011 | RSS ±0.0057**

**Try it yourself: Clearance fit limits**
Hole 2.0000/2.0010 in, shaft 1.9964/1.9980 in. Find the max and min clearance.
Handbook search: **clearance fit**

1. C_max = largest hole − smallest shaft = 2.0010 − 1.9964 = 0.0046 in.
2. C_min = smallest hole − largest shaft = 2.0000 − 1.9980 = 0.0020 in.

**Answer:** C_max = 0.0046 in; C_min = 0.0020 in

Check your answer: C_min > 0, so it is a clearance fit. A negative value would mean interference.



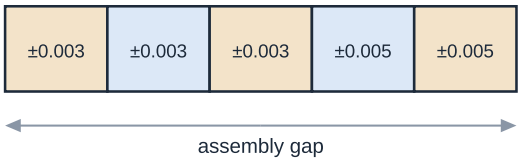

**C_max = 0.0046, C_min = 0.0020 in**

**Example 3: RSS stack**
Five parts: ±0.003, ±0.003, ±0.003, ±0.005, ±0.005 in. Find the worst-case and RSS tolerances.
Handbook search: **root sum square**

1. Worst case = Σt = ±0.019 in.
2. RSS = $\sqrt{\Sigma t^2}$ = ±0.0088 in.

**Answer:** WC ±0.019; RSS ±0.0088 in

Check your answer: RSS assumes independent, centered normal processes. It is not valid for very few parts or biased processes.




**worst case ±0.019 | RSS ±0.0088**

## 5. Common traps

- Swapping max and min clearance.
- Using RSS when the question asks for worst case.

# 6.B Fits and Tolerances: one-page cheat summary

- ▶ $C_{max} = H_{max} - s_{min}$
- ▶ $C_{min} = H_{min} - s_{max}$
- ▶ WC $\Sigma t$; RSS $\sqrt{\Sigma t^2}$
- ▶ FN = force fit

## Formulas

| | |
|---|---|
| Max clearance | $C_{max} = hole_{max} - shaft_{min}$ |
| Min clearance | $C_{min} = hole_{min} - shaft_{max}$ (negative = interference) |
| Worst case | $T = \Sigma t_i$ |
| RSS | $T = \sqrt{\Sigma t_i^2}$ |

## Search keywords

limits and fits, ANSI B4.1, ISO 286, tolerance stack-up RSS

## Top traps

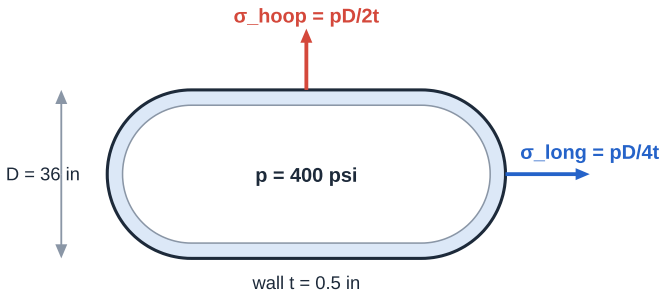
- Swapping max and min clearance.
- Using RSS when the question asks for worst case.

# 6.C Codes and Standards

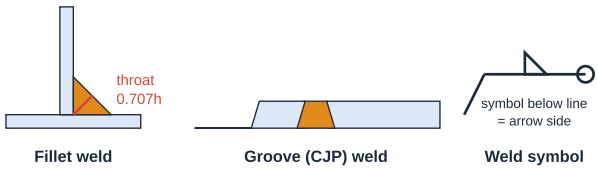
## 1. Key concepts

- Know which code covers what:
- ASME BPVC Section VIII: unfired pressure vessels. Section I: power boilers. ASME B31.1: power piping. B31.3: process piping.
- ASME Y14.5: GD&T. Y14.100: drawing practice. AWS A2.4: weld symbols. AWS D1.1: structural steel welding.
- AGMA 2001: gear rating. ANSI/ABMA 9/11 and ISO 281: bearing ratings. SAE J429 and ASTM A325/A490: bolt grades. ASTM E8: tension test. ISO 286 and ANSI B4.1: fits.
- The exam may supply excerpts of specific standards. Check the last page of the NCEES MDM specification for the current list.

## Diagrams




ASME BPVC Section VIII covers unfired pressure vessels.




AWS A2.4 defines welding symbols; AWS D1.1 covers structural steel welding.

## 2. Essential formulas

| Name | Formula | Variables / units |
|---|---|---|
| (no formulas) | Recognition topic. Know the scope of each code. | |

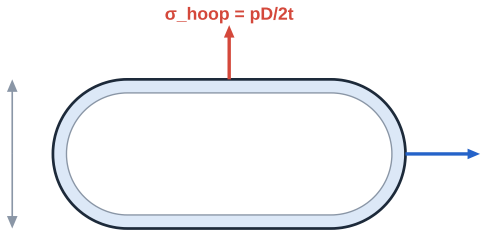
## 3. Handbook search keywords

ASME Section VIII | B31.3 | Y14.5 | AWS D1.1 | AGMA | ABMA

## 4. Worked examples

**Example 1: Identify the code**
You are designing process piping in a refinery. Which code governs?
Handbook search: **ASME B31.3**

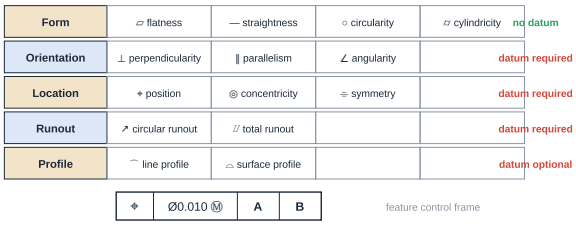
1. Process piping falls under ASME B31.3 (power-plant piping is B31.1).




**Answer:** ASME B31.3

Check your answer: Process piping → B31.3; power piping → B31.1; vessels → BPVC VIII.

**Try it yourself: Which standard for GD&T?**
A drawing uses feature control frames and datums. Which standard defines them?
Handbook search: **ASME Y14.5**

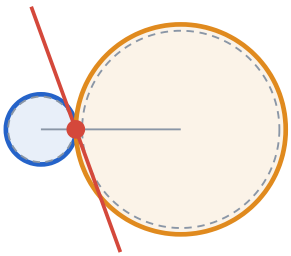



1. Dimensioning and tolerancing symbols come from ASME Y14.5.
2. ISO GPS (ISO 1101) is the international counterpart.

**Answer:** ASME Y14.5

Check your answer: Y14 is the drawing-standards family. B31 is piping and BPVC is boilers and vessels.

**Example 3: Gear rating standard**
You must rate spur gear bending and pitting strength for a US industrial drive. Which body publishes the method?
Handbook search: **AGMA**



1. AGMA (American Gear Manufacturers Association): AGMA 2001 for spur and helical gear rating.

**Answer:** AGMA 2001

Check your answer: The Lewis equation is the classic textbook start. AGMA adds geometry, dynamic, size and reliability factors.

## 5. Common traps

- Confusing B31.1 (power) with B31.3 (process).
- Confusing AWS A2.4 (symbols) with D1.1 (structural welding).

# 6.C Codes and Standards: one-page cheat summary

- ▶ BPVC VIII vessels
- ▶ B31.1 power / B31.3 process
- ▶ Y14.5 GD&T
- ▶ AWS D1.1 welding
- ▶ AGMA gears; ABMA bearings
- ▶ SAE J429 bolts

## Formulas

| | |
|---|---|
| (no formulas) | Recognition topic. Know the scope of each code. |

## Search keywords

ASME Section VIII, B31.3, Y14.5, AWS D1.1, AGMA, ABMA

## Top traps

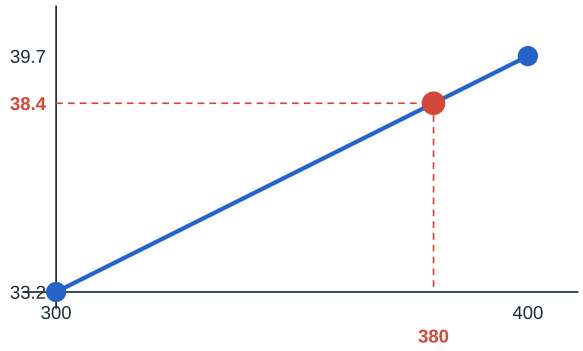
- Confusing B31.1 (power) with B31.3 (process).
- Confusing AWS A2.4 (symbols) with D1.1 (structural welding).

# 6.D Computational Methods

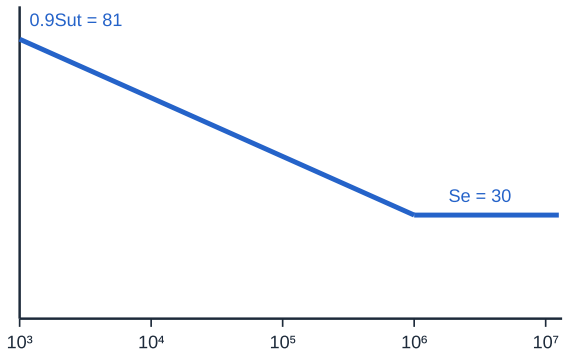
## 1. Key concepts

- Linear interpolation from property tables: $y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$.
- Root finding: Newton–Raphson $x_1 = x_0 - f(x_0)/f'(x_0)$. Bisection is slower but always converges.
- Numerical integration: trapezoid; Simpson's 1/3 rule (even number of intervals) is exact for cubics.
- FEA: refine the mesh where gradients are high. Sharp re-entrant corners are singularities, so model the real fillet. Check reactions against the hand-calculated loads.

## Diagrams




Linear interpolation between table rows.




Log-log fits turn power laws into straight lines.

## 2. Essential formulas

| Name | Formula | Variables / units |
|---|---|---|
| Interpolation | $y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$ | |
| Newton–Raphson | $x_{k+1} = x_k - f/f'$ | |
| Simpson 1/3 | $\int \approx (h/3)[f_0 + 4f_1 + 2f_2 + \ldots + f_n]$ | |
| Trapezoid | $\int \approx (h/2)[f_0 + 2f_1 + \ldots + f_n]$ | |

## 3. Handbook search keywords

linear interpolation · Newton-Raphson · Simpson's rule · finite element

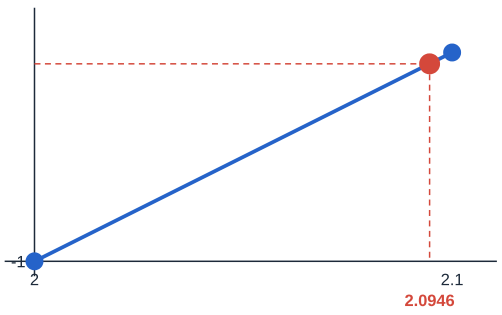
## 4. Worked examples

**Example 1: Newton–Raphson**
Find the root of $f(x) = x^3 - 2x - 5$ starting at $x_0 = 2$ (one iteration).
Handbook search: **Newton-Raphson**

1. $f(2) = -1$; $f'(x) = 3x^2 - 2 = 10$.
2. $x_1 = 2 - (-1)/10 = 2.1$.

**Answer:** 2.1 (true root ≈ 2.0946)

Check your answer: $f(x_1)$ should be much closer to zero than $f(x_0) = -1$.



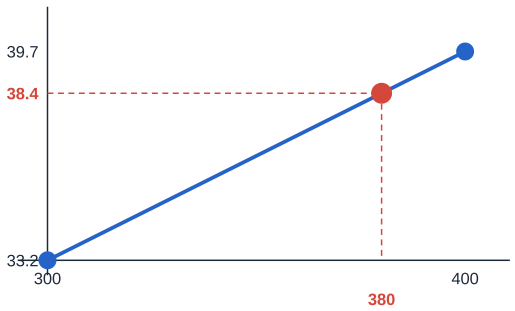
**Try it yourself: Linear interpolation**
A table lists 33.2 at 300 °F and 39.7 at 400 °F. Find the value at 380 °F.

Handbook search: **linear interpolation**

1. $y = y_1 + (x - x_1)/(x_2 - x_1)\cdot(y_2 - y_1) = 33.2 + 0.8 \times 6.5 = 38.4$.

**Answer:** 38.4

Check your answer: The answer must sit between 33.2 and 39.7, closer to 39.7.



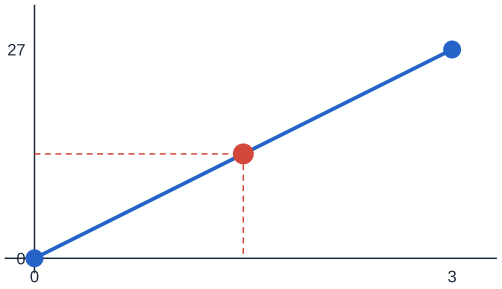

**Example 3: Simpson's 1/3 rule**
Integrate $x^3$ from 0 to 3 with 4 intervals.

Handbook search: **Simpson's rule**

1. h = 0.75; f = 0, 0.422, 3.375, 11.39, 27.
2. $I = h/3[f_0 + 4f_1 + 2f_2 + 4f_3 + f_4] = 20.2500$.

**Answer:** 20.25

Check your answer: Simpson is exact for cubics, so the answer equals the exact $3^4/4 = 20.25$.




## 5. Common traps

- Using Simpson's rule with an odd number of intervals.
- Inverting the slope in Newton's method.

# 6.D Computational Methods: one-page cheat summary

- ► Interp: $y_1 + \Delta y \cdot (x - x_1)/\Delta x$
- ► N–R: $x - f/f'$
- ► Simpson: $h/3[1\ 4\ 2\ 4\ \ldots 1]$
- ► FEA: singular corners do not converge

## Formulas

| | |
|---|---|
| Interpolation | $y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$ |
| Newton–Raphson | $x_{k+1} = x_k - f/f'$ |
| Simpson 1/3 | $\int \approx (h/3)[f_0 + 4f_1 + 2f_2 + \ldots + f_n]$ |
| Trapezoid | $\int \approx (h/2)[f_0 + 2f_1 + \ldots + f_n]$ |

## Search keywords

linear interpolation, Newton-Raphson, Simpson's rule, finite element

## Top traps

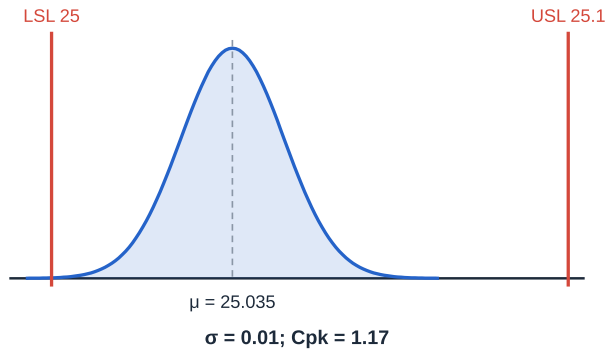
- Using Simpson's rule with an odd number of intervals.
- Inverting the slope in Newton's method.

# 6.E Instrumentation, Testing, Inspection, and Quality

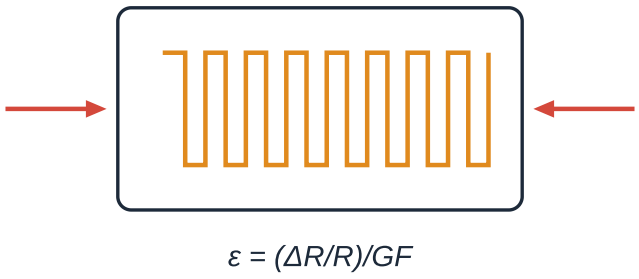
## 1. Key concepts

- Measurement: the instrument resolution should be about 1/10 of the tolerance. A micrometer reads 0.0001 in; a caliper reads 0.001 in.
- Strain gage: $\varepsilon = (\Delta R/R)/GF$.
- Process capability: $C_p = (USL - LSL)/6\sigma$; $C_{pk} = \min(USL - \mu, \mu - LSL)/3\sigma$. 1.33 or more is good.
- Control charts: $\bar{X}$ chart limits $\bar{\bar{X}} \pm A_2\bar{R}$; R chart limits $D_3\bar{R}$ and $D_4\bar{R}$.
- NDT: UT and RT find subsurface flaws. Dye penetrant finds surface-breaking flaws only. Magnetic particle finds surface and near-surface flaws (ferromagnetic parts only).

## Diagrams




Process capability: Cp and Cpk.




Quarter-bridge strain gage.

## 2. Essential formulas

| Name | Formula | Variables / units |
|---|---|---|
| Strain gage | $\varepsilon = (\Delta R/R)/GF$ | |
| Cp | $C_p = (USL - LSL)/(6\sigma)$ | |
| Cpk | $C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$ | |
| $\bar{X}$ chart | $UCL = \bar{\bar{X}} + A_2\bar{R}$ | |

## 3. Handbook search keywords

process capability Cpk · control chart · strain gage factor · nondestructive testing

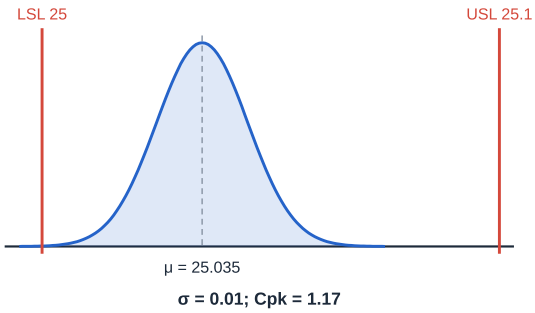
## 4. Worked examples

**Example 1: Cpk**
Limits 25.00–25.10 mm, μ = 25.035, σ = 0.010. Find $C_p$ and $C_{pk}$.
Handbook search: **process capability**

1. $C_p = 0.10/0.06 = 1.67$.
2. $C_{pk} = \min(0.065, 0.035)/0.03 = 1.17$ (not centered).

**Answer:** $C_p = 1.67$, $C_{pk} = 1.17$

Check your answer: Cpk ≤ Cp always. They are equal only when the process is centered.

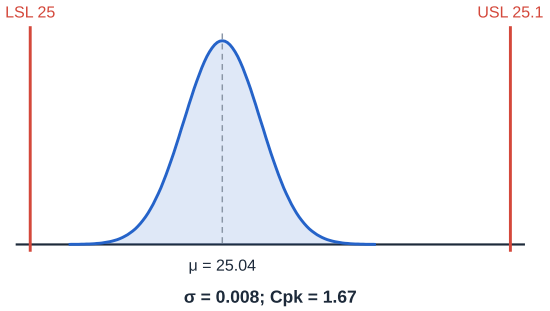


**Try it yourself: Cpk**
Limits 25.00–25.10 mm, μ = 25.040, σ = 0.008 mm. Find Cpk.
Handbook search: **process capability**

1. Distance to USL = 0.060; to LSL = 0.040.
2. Cpk = min/3σ = 0.040/0.024 = 1.67.

**Answer:** 1.67

Check your answer: Cpk ≥ 1.33 is a common target. This process meets it.




**Example 3: Strain from a gage**
GF = 2.0, ΔR = 0.24 Ω on a 120 Ω gage. Find the strain.
Handbook search: **strain gage**

1. ε = (ΔR/R)/GF = (0.24/120)/2.0 = 0.0010 = 1000 με.

**Answer:** 1000 με

Check your answer: 1,000 με in steel is about 29 ksi, a reasonable working stress.




## 5. Common traps

- Using C_p when the process is off center.
- Expecting dye penetrant to find internal flaws.

# 6.E Instrumentation, Testing, Inspection, and Quality: one-page cheat summary

- ► $\varepsilon = (\Delta R/R)/GF$
- ► $C_{pk} = \min(\ldots)/3\sigma$
- ► $UCL = \bar{\bar{X}} + A_2\bar{R}$
- ► UT/RT subsurface; PT surface only

## Formulas

| | |
|---|---|
| Strain gage | $\varepsilon = (\Delta R/R)/GF$ |
| Cp | $C_p = (USL - LSL)/(6\sigma)$ |
| Cpk | $C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$ |
| $\bar{X}$ chart | $UCL = \bar{\bar{X}} + A_2\bar{R}$ |

## Search keywords

process capability Cpk, control chart, strain gage factor, nondestructive testing

## Top traps

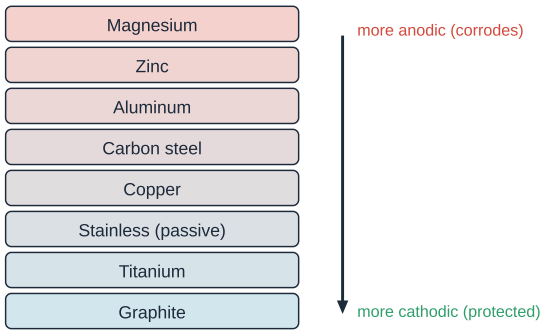
- Using C_p when the process is off center.
- Expecting dye penetrant to find internal flaws.

# 6.F Chemical Processes (Corrosion, Coatings, Treatments)

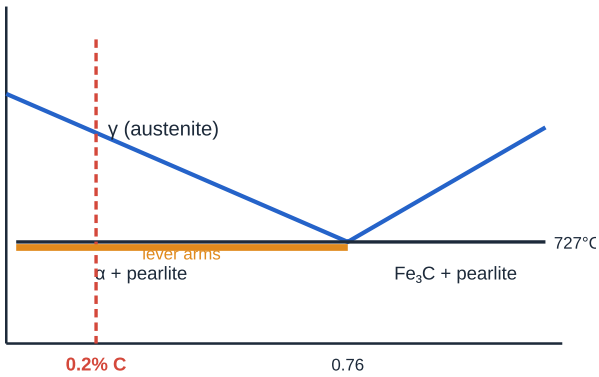
## 1. Key concepts

- Galvanic corrosion: the more anodic metal corrodes (Mg, Zn, Al, steel, … Cu, stainless (passive), Ti, graphite). Avoid a small anode with a large cathode; isolate dissimilar metals.
- Protection: coatings, galvanizing (zinc as a sacrificial anode), cathodic protection, inhibitors, material selection.
- Hydrogen embrittlement: high-strength steels after plating or pickling. Bake after plating.
- Surface treatments: carburizing and nitriding (hard case), shot peening (compressive residual stress), anodizing (aluminum).

## Diagrams




Galvanic series in seawater: anodic metals corrode.




Heat treatment and surface hardening start from the Fe–C diagram.

## 2. Essential formulas

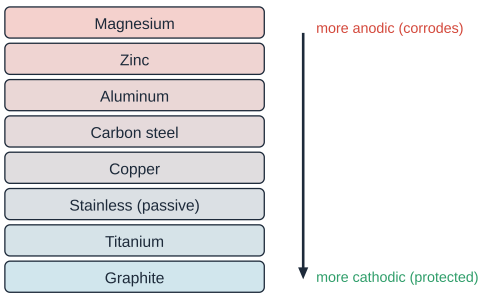
| Name | Formula | Variables / units |
|---|---|---|
| (mostly concept) | Galvanic series order; area ratio effect | |

## 3. Handbook search keywords

galvanic series · corrosion protection · hydrogen embrittlement · surface treatment

## 4. Worked examples

**Example 1: Galvanic pair**
Aluminum rivets in a stainless panel in seawater. What happens and how do you fix it?
Handbook search: **galvanic corrosion**

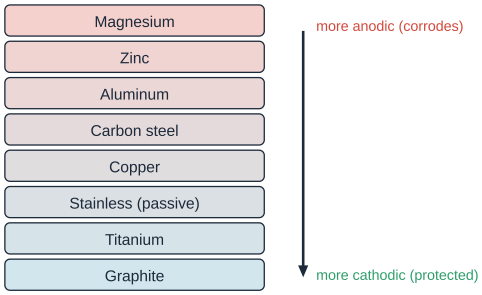



1. Aluminum is anodic and the rivet area is small: severe rivet corrosion.
2. Fix: use stainless rivets (large anode, small cathode), isolate the metals, or coat the cathode.

**Answer:** The rivets corrode; change the material or isolate

Check your answer: Small anode with large cathode is the worst case. Swap it around or isolate the metals.

**Try it yourself: Steel bolt in copper**
A steel bolt fastens copper sheet in seawater. What happens?
Handbook search: **galvanic series**

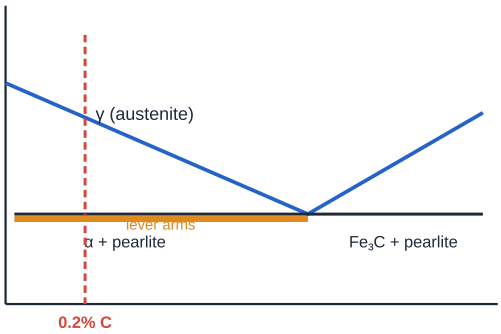



1. Steel is anodic to copper in the galvanic series.
2. The small steel anode with a large copper cathode corrodes quickly.
3. Fix: use a compatible fastener (e.g. bronze or Monel), insulating washers, or coat the cathode.

**Answer:** The steel bolt corrodes rapidly

Check your answer: Ask which metal is higher (more anodic) on the chart, then compare areas.

**Example 3: Carburizing depth**
A low-carbon (0.2% C) gear is carburized and quenched. Why use low-carbon steel?
Handbook search: **carburizing**




1. Low-C core stays tough after quenching.
2. The diffused high-C case (~0.8% C) becomes hard martensite for wear resistance.
3. Compressive residual stress in the case also improves fatigue life.

**Answer:** Hard case plus tough core

Check your answer: If the whole part were 0.8% C, it would be hard but brittle through the section.

## 5. Common traps

- Coating only the anode (any holiday then corrodes fast). Coat the cathode, or both.

# 6.F Chemical Processes (Corrosion, Coatings, Treatments): one-page cheat summary

- ► Anode corrodes
- ► Small anode/large cathode = bad
- ► Zinc protects steel sacrificially
- ► Bake after plating

## Formulas

| | |
|---|---|
| (mostly concept) | Galvanic series order; area ratio effect |

## Search keywords

galvanic series, corrosion protection, hydrogen embrittlement, surface treatment

## Top traps

- Coating only the anode (any holiday then corrodes fast). Coat the cathode, or both.